CLINICAL · APPROACH TO

Approach to a lump in the neck

A lump in the neck is one of the commonest reasons to see an ENT surgeon, and one of the most dangerous to mismanage. In a child it is almost always a reactive node or a congenital cyst. In an adult over forty who smokes, a persistent lateral lump is a metastatic squamous carcinoma until proved otherwise. The rule is simple: a full head and neck examination with endoscopy, imaging, and fine-needle aspiration come before any open biopsy. A node cut out casually seeds the neck and spoils the cure.

AT A GLANCE

- Adult neck lump is cancer until shown otherwise.
- Scope, FNA, image.
- No open biopsy first.
- A cystic node in middle age may be HPV-related.

1 CHAPTER ONE Approach

Where, how old, how long.

1.1 The age rule

In children and young adults, inflammatory and congenital masses dominate, and lymphoma and thyroid cancer are the malignancies to remember. In adults over about forty, neoplasms dominate, and most are metastases, usually from squamous carcinoma of the upper aerodigestive tract. The old "rule of 80" puts it memorably: about 80% of non-thyroid neck masses in adults are neoplastic, 80% of those are malignant, and 80% of the malignant ones are metastatic, mostly from a primary above the clavicles.

1.2 The time rule

Duration is almost as informative as age. A mass present for days is usually inflammatory, one present for months neoplastic, and one present for years

congenital or benign.

1.3 Site

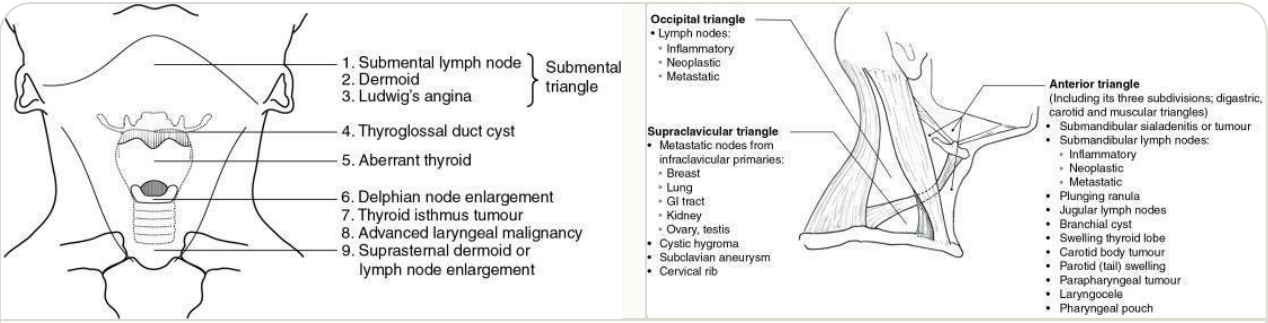


FIGURE 1 First, swellings in the midline of the neck. Second, lateral swellings grouped by the triangles of the neck.

Dhingra, figs 77.1 and 77.2

NECK MASSES BY SITE

SITE	COMMON CAUSES
Midline, above the hyoid	Submental nodes, sublingual dermoid, thyroglossal cyst (suprahyoid), plunging ranula
Midline, below the hyoid	Thyroglossal cyst (commonest at the hyoid), thyroid isthmus, Delphian and pretracheal nodes, laryngocele, suprasternal dermoid, thymic cyst
Submandibular triangle	Submandibular nodes and gland (sialadenitis, stone, tumour), plunging ranula
Carotid triangle	Jugulodigastric and deep cervical nodes, branchial cyst, carotid body tumour, vagal schwannoma, laryngocele, carotid aneurysm
Parotid region and angle of the jaw	Parotid tumour or node, parapharyngeal mass, sebaceous cyst
Posterior triangle	Accessory chain nodes (nasopharynx, lymphoma, tuberculosis), cystic hygroma, lipoma, neural tumours, cervical rib
Supraclavicular	Virchow node (stomach and other infraclavicular primaries), lung and breast metastases, lymphoma, tuberculosis
Within sternomastoid	Sternomastoid tumour of infancy

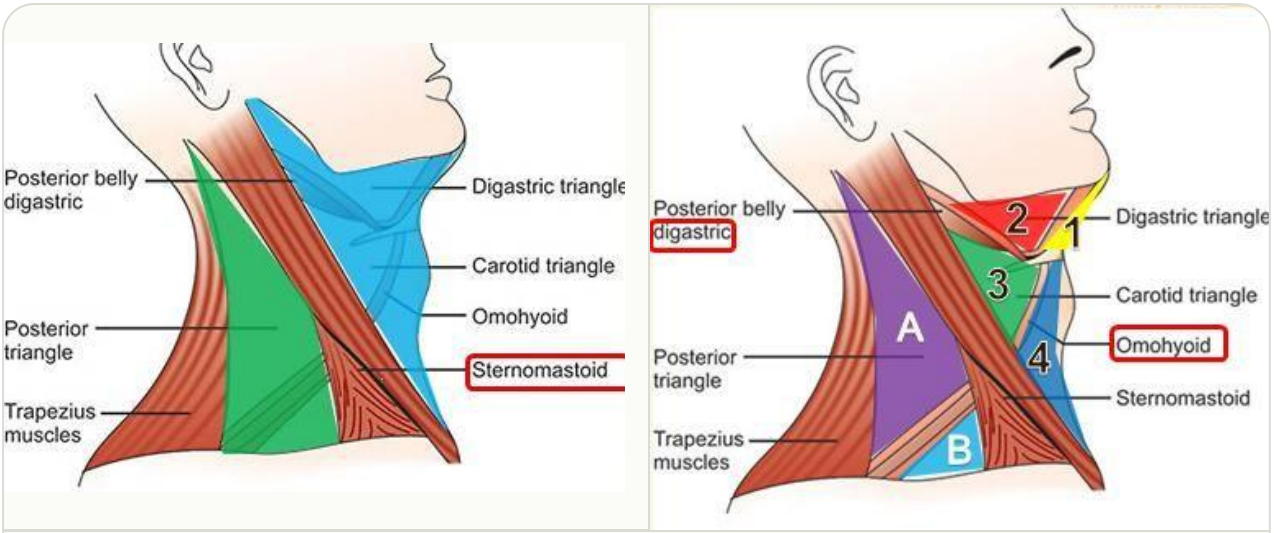


FIGURE 2 The triangles of the neck and their subdivisions.
Anatomy atlas · ENT Steps p. 270

CHAPTER TWO

2 Lymph nodes

Levels and drainage.

2.1 The levels

NECK NODE LEVELS

LEVEL	BOUNDARIES	DRAINS
IA (submental)	Between the anterior bellies of digastric, above the hyoid	Lower lip, chin, floor of mouth, tongue tip
IB (submandibular)	Digastric bellies and the mandible	Oral cavity, anterior nasal cavity, face, submandibular gland
II (upper jugular)	Skull base to the hyoid, around the internal jugular vein. IIA in front of the accessory nerve, IIB behind it	Oral cavity, oropharynx, nasopharynx, larynx, parotid
III (middle jugular)	Hyoid to the cricoid	Oropharynx, hypopharynx, larynx
IV (lower jugular)	Cricoid to the clavicle	Hypopharynx, larynx, thyroid, cervical oesophagus

LEVEL	BOUNDARIES	DRAINS
V (posterior triangle)	Behind sternomastoid, in front of trapezius. VA above the cricoid, VB below	Nasopharynx, oropharynx, scalp and skin, thyroid
VI (central)	Hyoid to the sternal notch, between the carotids	Thyroid, larynx (subglottis), hypopharynx, cervical oesophagus
VII (superior mediastinal)	Below the sternal notch	Thyroid, oesophagus

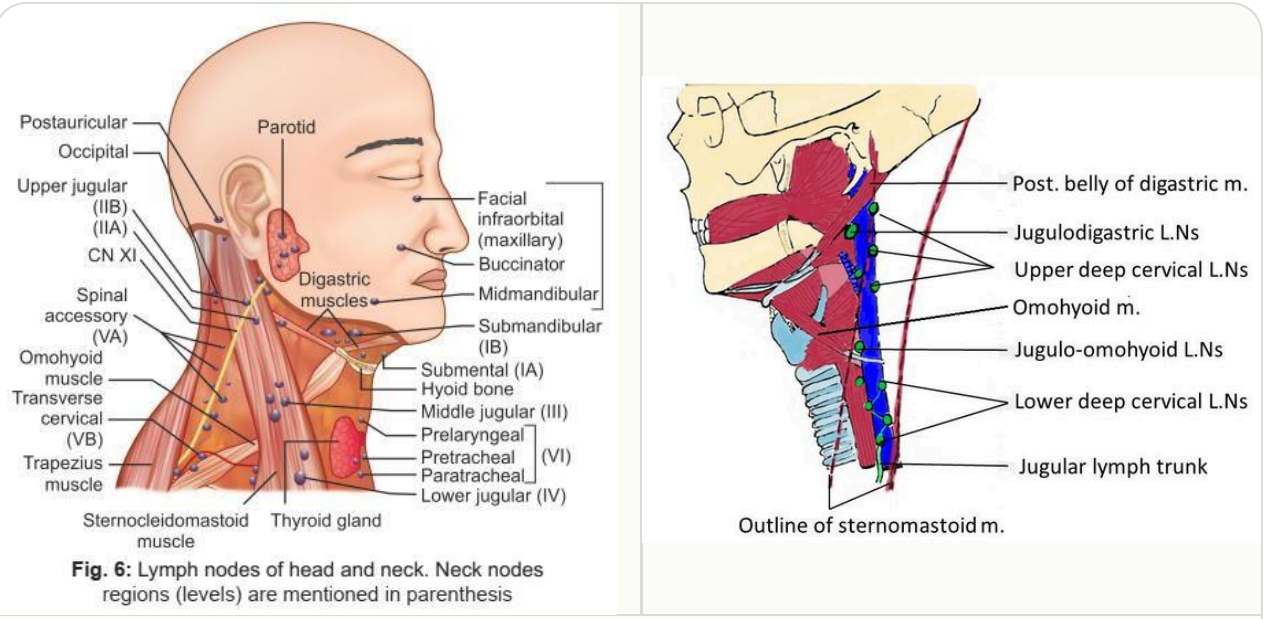


FIGURE 3 The lymph node levels of the neck, and the deep cervical chain.

Anatomy atlas · ENT Steps p. 285, Doss p. 198

READING THE NODE BACK TO THE PRIMARY

- **Jugulodigastric node (level II):** tonsil, tongue base, nasopharynx, supraglottis.
- **Posterior triangle nodes:** suggest the nasopharynx.
- **Level VI:** suggests the thyroid.
- **Left supraclavicular node (Virchow, *Troisier sign*):** suggests an abdominal or thoracic primary.
- **Cystic level II node in an adult:** a metastasis from *HPV-positive oropharyngeal carcinoma* (tonsil or tongue base) or from papillary thyroid carcinoma, not a "branchial cyst", until proved otherwise.

3 CHAPTER THREE Assessment

History, examination, tests.

3.1 History

Ask how long the lump has been present, whether it is growing or painful, and whether it changes with meals or infections. Then ask about the upper aerodigestive tract: hoarseness, dysphagia, odynophagia, referred otalgia, nasal obstruction or bleeding, a unilateral serous otitis, and weight loss. Record the risk factors (tobacco, alcohol and betel quid; sexual history for HPV; exposure to tuberculosis; HIV risk; contact with cats for *Bartonella* or toxoplasmosis; travel; and previous skin cancer or radiation) and any "B" symptoms of fever, night sweats and weight loss that suggest lymphoma or tuberculosis.

3.2 Examination

Palpate the neck from behind, level by level, noting the size, consistency, fixity, pulsation and transillumination of the lump and whether it moves with swallowing (thyroid, thyroglossal cyst) or with tongue protrusion (thyroglossal cyst). Then examine the whole head and neck: scalp, skin, ears, nose and oral cavity, palpate the tonsils and tongue base, and look at the nasopharynx, larynx and hypopharynx with a flexible nasendoscope. Finish with the axillae, groins, liver and spleen.

3.3 Investigation

1. **Ultrasound with FNA** is the first test for most adult masses. Ultrasound shows whether the mass is cystic or solid, the node architecture and its vascularity; FNA gives cytology, with p16/HPV, EBV, thyroglobulin, flow cytometry for lymphoma, and culture and acid-fast stains as needed. A non-diagnostic aspirate is repeated or replaced by a core biopsy.
2. **Contrast CT or MRI of the neck** shows the extent and may reveal a primary.
3. **PET-CT** is used for an unknown primary, done *before* panendoscopy and biopsies, and for staging and distant disease.
4. **Chest imaging.**

5. **Blood tests:** count, ESR, serology for EBV, HIV, toxoplasma and *Bartonella*, and tests for tuberculosis.
6. **Panendoscopy under anaesthesia** with directed biopsies, tonsillectomy and tongue-base mucosectomy to find an occult primary.

DO NOT EXCISE AN ADULT NECK NODE FIRST

An open biopsy of a node that turns out to be metastatic squamous carcinoma breaches the fascial planes and contaminates the field. It raises rates of local recurrence and complications. If open biopsy is unavoidable after negative FNA and endoscopy (for example, to type a suspected lymphoma), plan it so that the incision can be included in a neck dissection, with frozen section, and be ready to proceed to dissection.

OPERATIVE GUIDES IN THEATRE 2 operations

Fine-needle aspiration, and management of the unknown primary.

Office-Based Procedures: Fine-Needle Aspiration Biopsy of a Mass in the Head and Neck
FNA

Management of the Unknown Primary Cancer
Unknown primary

4 CHAPTER FOUR Inflammatory causes

The commonest masses.

4.1 Reactive lymphadenitis

Reactive nodes are tender, mobile and often bilateral, and accompany a sore throat, a dental infection or a skin infection. They settle within two to six weeks. A node that persists beyond that, grows, is hard, or lies in the supraclavicular fossa needs investigation.

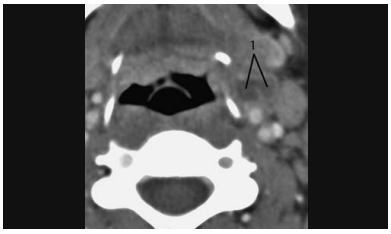
4.2 Acute suppurative lymphadenitis

Suppurative lymphadenitis, caused by *Staphylococcus aureus* and group A streptococcus, mainly affects young children. The node is hot and tender and may

form an abscess. It is treated with antibiotics, and an abscess is drained by aspiration or incision. A neck abscess in an adult should raise the suspicion of a necrotic metastasis or tuberculosis.

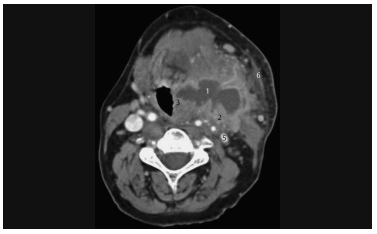
IMAGING IN THE APP 2 cases

Cervical lymphadenitis and neck abscess.



Cervical Lymphadenitis

Infection & inflammation · 9 images



Abscess of the Neck

Infection & inflammation · 1 image

4.3 Tuberculous lymphadenitis (scrofula)

Tuberculous lymphadenitis is very common in endemic areas and more frequent with HIV. It affects any age, usually the deep cervical and posterior triangle nodes, and passes through recognisable stages: discrete nodes, then matted nodes with periadenitis, then caseation and a *cold abscess*, then a "collar-stud" abscess through the deep fascia, and finally sinus formation.

The diagnosis rests on FNA or biopsy showing caseating granulomas, acid-fast bacilli on Ziehl–Neelsen staining, culture and GeneXpert with rifampicin resistance testing, supported by a chest radiograph, a tuberculin test or interferon-gamma release assay, and an HIV test. Treatment is standard antituberculous therapy: isoniazid, rifampicin, pyrazinamide and ethambutol for two months, then isoniazid and rifampicin for four. Nodes may enlarge temporarily during treatment, a *paradoxical reaction*. Surgery is reserved for drug failure or residual masses, and abscesses are aspirated rather than incised.

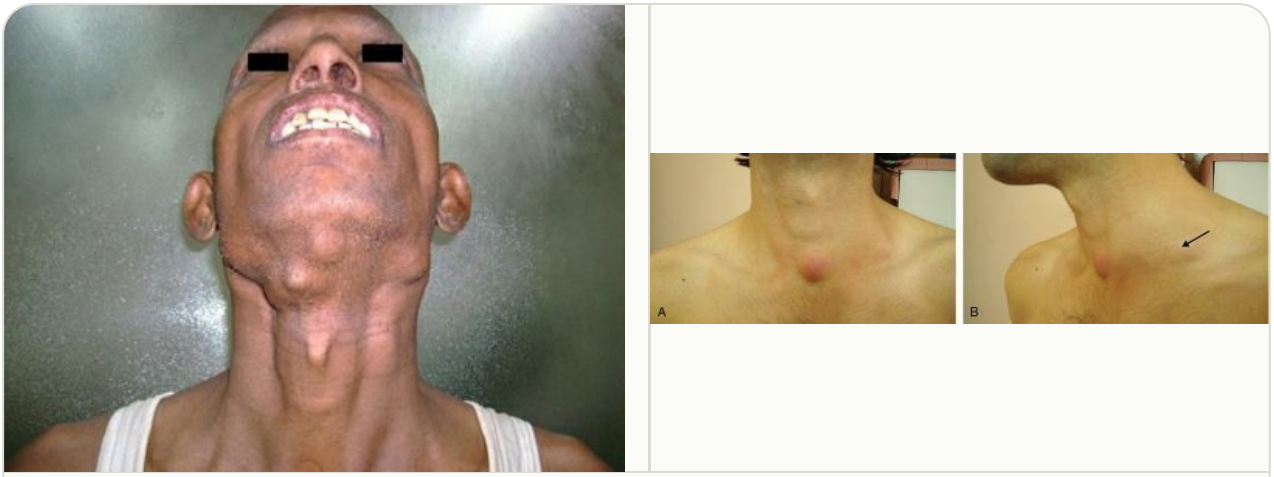


FIGURE 4 First, multiple tuberculous nodes in the neck. Second, (A) a caseating suprasternal node and (B) supraclavicular tuberculous nodes.
Dhingra, figs 77.8 and 77.9

4.4 Atypical mycobacteria

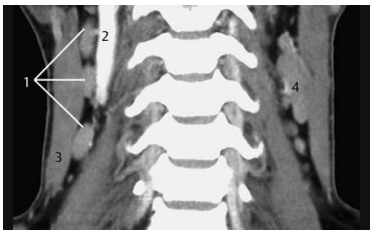
Atypical mycobacterial infection affects children aged one to five as a painless node, usually submandibular or preauricular, with violaceous, thin overlying skin and a tendency to form sinuses. The chest film is normal and the tuberculin reaction weak. Complete excision is the treatment, or curettage where the facial nerve is at risk, with clarithromycin as an alternative; incision and drainage leaves a chronic sinus.

4.5 Other infections

Cat-scratch disease from *Bartonella henselae* causes a tender, often suppurating node after a cat scratch; it is diagnosed by serology and treated with azithromycin when severe. **Toxoplasmosis** causes posterior cervical nodes in young adults and is diagnosed by serology. **Infectious mononucleosis** from EBV gives generalised nodes, exudative tonsillitis and splenomegaly. **HIV** causes persistent generalised lymphadenopathy, and its nodes may harbour tuberculosis, lymphoma or Kaposi sarcoma. **Actinomycosis** produces a woody swelling with sinuses near the jaw.

IMAGING IN THE APP 1 case

Cervical lymphadenopathy in HIV.



Lymphadenopathy, HIV

Infection & inflammation · 1 image

4.6 Non-infective inflammatory causes

Sarcoidosis produces non-caseating granulomas. **Kikuchi–Fujimoto disease**, histiocytic necrotising lymphadenitis, affects young women with tender nodes and fever and settles on its own. **Kimura disease** affects young Asian men with a preauricular mass, eosinophilia and a high IgE. **Castleman disease** is unicentric or multicentric, and **Rosai–Dorfman disease** causes massive painless nodes.

5 CHAPTER FIVE Neoplastic causes

Metastases and lymphoma.

5.1 Metastatic squamous carcinoma

The typical patient has a hard, painless, enlarging node, often in level II or III. He may be an older smoker and drinker, or a younger non-smoker with HPV-positive oropharyngeal cancer, whose nodes are often cystic. A node fixed to deep structures or to the skin means extranodal spread. The primary is sought in the tonsil, tongue base, nasopharynx, pyriform sinus, larynx and oral cavity, by examination, endoscopy, imaging and panendoscopy. Supraclavicular nodes may come from below the clavicle: from the lung, breast, oesophagus, stomach, pancreas, kidney, testis or ovary.

5.2 Carcinoma of unknown primary

When a squamous metastasis has no primary on examination, endoscopy and imaging, the search proceeds in a set order:

1. PET-CT.
2. Panendoscopy with biopsies.
3. Ipsilateral (or bilateral) tonsillectomy and transoral tongue-base mucosectomy.
4. p16 and EBV staining of the node, which point to the oropharynx or nasopharynx.

Treatment is neck dissection with radiotherapy to the neck and the likely mucosal sites, or primary chemoradiotherapy. The outcome is better than many expect, especially with HPV-positive disease.

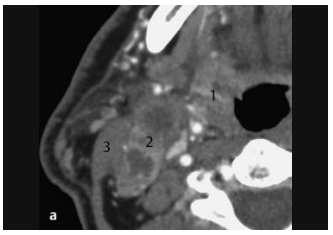
5.3 Lymphoma

Hodgkin lymphoma affects young adults, with rubbery, painless, discrete nodes, often low in the neck or supraclavicular, sometimes with a mediastinal mass, B symptoms and pain after alcohol; the diagnostic cell is the Reed–Sternberg cell. **Non-Hodgkin lymphoma** affects older adults and involves both nodes and extranodal sites, including Waldeyer's ring (tonsil, nasopharynx, tongue base), the thyroid, the salivary glands and the sinonasal tract, where NK/T-cell lymphoma occurs.

FNA with flow cytometry may suggest lymphoma, but a *core or excision biopsy* is usually needed to classify it, and fresh tissue must be sent. Treatment follows staging, with chemotherapy with or without radiotherapy.

IMAGING IN THE APP 1 case

Non-Hodgkin lymphoma in the neck.



Lymphoma, Non-Hodgkin
Tumours · 13 images

5.4 Other neoplasms

Thyroid carcinoma, including cystic nodal metastases of papillary carcinoma, salivary tumours, paragangliomas and schwannomas, lipomas and, rarely, sarcomas complete the neoplastic list.

6 CHAPTER SIX Other masses

Not nodes.

6.1 Swellings that are not nodes

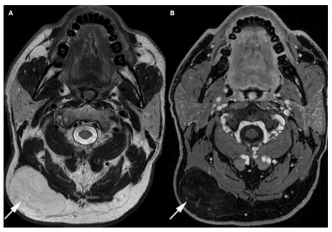
A **laryngocele** is an air-filled dilatation of the laryngeal saccule. An *external* laryngocele pierces the thyrohyoid membrane to form a lateral neck swelling that enlarges on a Valsalva manoeuvre, coughing or blowing, as in trumpeters and glassblowers; infection turns it into a *laryngopyocele*, and a ventricular or supraglottic carcinoma obstructing the saccule must always be excluded. A **pharyngeal pouch** forms a soft swelling low in the left neck that gurgles, with regurgitation. A **plunging ranula** is a soft, transilluminant submandibular swelling from the sublingual gland, cured by removing that gland.

A **cervical rib** is a hard supraclavicular lump with thoracic outlet symptoms: arm paraesthesia from pressure on the lower trunk of the brachial plexus, and coldness, claudication, aneurysm or emboli from compression of the subclavian artery. It is excised if it causes symptoms. A **lipoma** is soft and lobulated with a slipping edge, commonly in the posterior triangle and nape, and **sebaceous and epidermoid cysts**

lie in the skin with a punctum. Finally, a **carotid aneurysm**, or an ectatic, tortuous carotid in an elderly hypertensive patient, can present as a pulsatile lump.

IMAGING IN THE APP 1 case

Lipoma of the neck.



Lipoma of the neck

Tumours · 1 image

7 CHAPTER SEVEN Children

A different list.

7.1 Neck lumps in children

Most neck lumps in children are reactive lymph nodes, which are common, small, mobile and settle within weeks. Congenital lesions follow: **thyroglossal duct cysts** in the midline, moving with tongue protrusion; **branchial cysts, sinuses and fistulae**, most from the second arch along the anterior border of sternocleidomastoid; **dermoid cysts**; **lymphatic malformations** (cystic hygroma), which are soft, transilluminate and may respond to sclerotherapy; and **haemangiomas**. Infection gives acute suppurative lymphadenitis and abscess, and **non-tuberculous mycobacterial** infection gives a violaceous, painless node in a toddler, often in the submandibular region, which is best treated by excision when feasible.

Malignancy is uncommon but must be considered. Features that call for urgent assessment are a node larger than 2 cm or growing, a hard, fixed or matted node, a supraclavicular node, nodes persisting beyond six weeks, and systemic symptoms

(fever, night sweats, weight loss, bruising, pallor, hepatosplenomegaly). Lymphoma is the commonest head and neck malignancy in children, followed by rhabdomyosarcoma, neuroblastoma and thyroid cancer. Ultrasound is the first test, avoiding radiation; a chest X-ray excludes a mediastinal mass before any anaesthetic, and an excision biopsy is often needed for diagnosis.

8 CHAPTER EIGHT Pathway

Lump to diagnosis.

THE ADULT NECK LUMP

1 Clinical picture

→ **acute, tender, with infection**
antibiotics and review in two to three weeks

→ **midline moving with swallowing**
thyroid pathway

→ **any red flag or persisting over three weeks**
step 2

2 Full head and neck examination

→ flexible endoscopy, skin, oral cavity, tonsils and tongue base

→ **obvious primary**
biopsy it and stage

3 Ultrasound with FNA or core

→ **squamous carcinoma**
CT or MRI, PET-CT, panendoscopy, tonsillectomy and tongue-base mucosectomy if no primary

→ **lymphoma suspected**
core biopsy or planned excision for flow cytometry

→ **thyroid**
thyroid pathway

→ **tuberculosis**
culture, treat

4 **Still no diagnosis**

→ repeat core biopsy

→ open biopsy planned for possible neck dissection, with frozen section

9 CHAPTER NINE Neck dissection

The principle.

9.1 Types and indications

Neck dissection removes the lymph nodes and fat of the neck en bloc. A **radical** dissection removes levels I to V with the accessory nerve, internal jugular vein and sternomastoid; a **modified radical** dissection removes the same levels while preserving one or more of those structures; a **selective** dissection removes only the levels at risk, such as the supraomohyoid dissection (I to III) for the oral cavity and the lateral dissection (II to IV) for the larynx and pharynx; and an **extended** dissection removes additional nodes or structures. A **therapeutic** dissection treats clinically positive nodes, while an **elective** dissection treats a clinically negative neck when the risk of occult metastases exceeds about 15 to 20%.

OPERATIVE GUIDES IN THEATRE 3 operations

Selective, modified radical and radical neck dissection.

Selective Neck Dissection
Selective neck dissection

Modified Radical Neck Dissection
Modified radical

Radical Neck Dissection
Radical

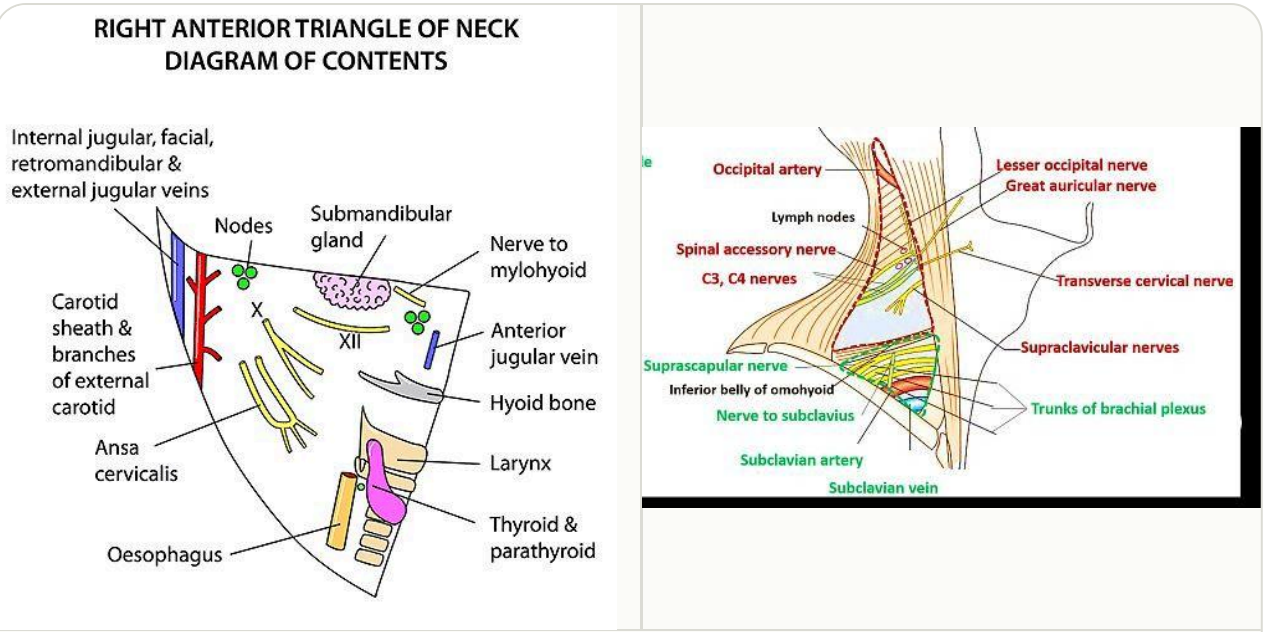


FIGURE 5 Contents of the anterior and posterior triangles.

Anatomy atlas · ENT Steps p. 271, ENT Steps p. 272

10 CHAPTER TEN For the examination

What the examiner wants to hear.

10.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Neck lumps in children** and the features that call for biopsy.
- **A systematic approach:** age, duration and site.
- **Node levels and drainage;** Virchow node.
- **The work-up of an adult neck mass:** full examination and endoscopy, then ultrasound and FNA, then imaging, then panendoscopy. Why open biopsy is avoided.
- **Unknown primary:** PET-CT before panendoscopy, tonsillectomy and tongue-base mucosectomy, p16 and EBV.
- **Tuberculous lymphadenitis:** stages, diagnosis and treatment.
- **Lymphoma:** needs tissue for typing.

10.2 Common pitfalls

COMMON PITFALLS

- Calling a cystic level II mass in an adult a branchial cyst.
- Excision biopsy of a node before examining the upper aerodigestive tract.
- Incising a tuberculous or atypical mycobacterial abscess, which leaves a chronic sinus.
- Missing the nasopharynx in a patient with posterior triangle nodes and a unilateral effusion.

Written for this app from Oxford Textbook of Otolaryngology ch. 50, 67, Oxford Specialist Handbook ch. 3, 12, Oxford Handbook of ENT ch. 3, 14, Dhingra ch. 77 and K. J. Lee ch. 11, 38, 40. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Cervical Lymphadenitis

IMAGING FINDINGS

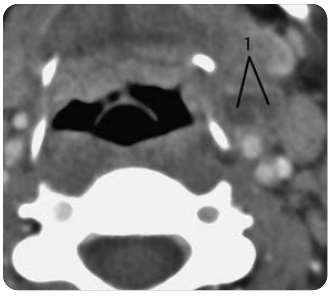
- Enlarged nodes with preserved fatty hilum (reactive)
- Suppurative node: thick rim around a fluid centre with surrounding inflammation
- Matted necrotic nodes in tuberculosis

TOP DIFFERENTIALS

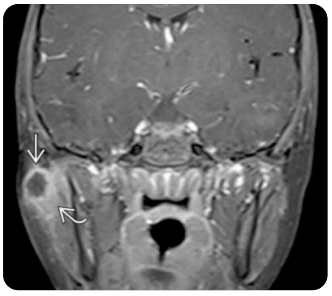
- Metastatic squamous carcinoma
- Second branchial cleft cyst
- Atypical mycobacteria

CHECKLIST

- Ultrasound first in children
- Progression to abscess



CT with intravenous contrast enhancement, axial. Various enlarged level III lymph nodes are present near the internal carotid artery and jugular…



Coronal T1 C+ FS MR in an 8-year-old boy with right cheek swelling shows a peripherally enhancing, centrally necrotic intraparotid mass with…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 314.

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Abscess of the Neck

IMAGING FINDINGS

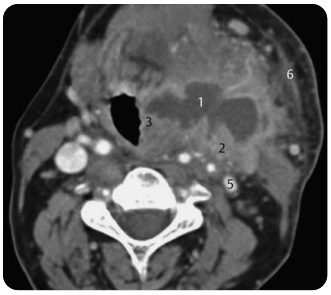
- Low-density collection with a well-formed enhancing rim
- Surrounding cellulitis and fat stranding
- The source: tooth, tonsil, node, stone or foreign body

TOP DIFFERENTIALS

- Phlegmon (no fluid centre)
- Necrotic metastatic node
- Infected congenital cyst

CHECKLIST

- Airway, mediastinum and jugular vein
- Which space: decides the drainage route



CT with intravenous contrast enhancement, axial.A large abscess (1) with a thick, contrast-enhanced capsule (2). The abscess is extendin…

NECK: TRIANGLES, FASCIA & NODES · TUMOURS

Lymphoma, Non-Hodgkin

IMAGING FINDINGS

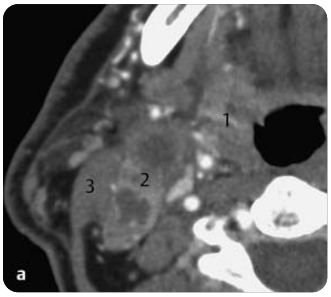
- Large, homogeneous, non-necrotic nodes in many chains
- Waldeyer ring involvement
- Strong diffusion restriction

TOP DIFFERENTIALS

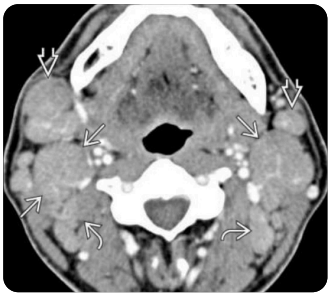
- Reactive nodes
- Tuberculosis
- Metastatic carcinoma

CHECKLIST

- Extranodal sites: tonsils, sinuses, orbit, thyroid
- Core or excision biopsy, not FNA alone



CT with intravenous contrast enhancement, axial. On this slice, an area of different structure with enhancement is seen at the base of…



Axial CECT shows multiple large, round, solid lymph nodes in every chain of the suprahyoid neck. Bilateral nodes are evident in levels IIA, IIB, and…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 336.

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Lymphadenopathy, HIV

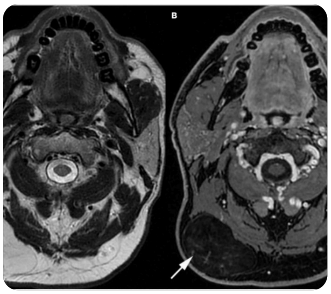
- HIV infections may be associated with coexisting lymphoma and chronic lymphadenitis. Coexisting diseases may have a different clinical presentation in HIV.



CT with intravenous contrast enhancement, coronal. Serology revealed HIV infection. On the right side, multiple homogeneously…

NECK: TRIANGLES, FASCIA & NODES · TUMOURS

Lipoma of the neck



Axial T2W (a) and fat suppressed T1 post-contrast (b) MR images show a lipomatous mass (arrows) in the right posterior neck with no complex…